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(UGC AUTONOMOUS INSTITUTION)
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SRS REVIEW

Autism Spectrum Disorder Analysis for Toddlers

Batch No : B20

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Introduction to SRS

- **Software Requirement Specification** (SRS) outlines system functionality and constraints.
- Defines **scope** , **system environment**, and **performance requirements**.
- System analysis plays **a crucial role** in shaping the development process by identifying what the system should do (functional requirements).
- The Autism Screening System is a **data-driven web application** built using machine learning, and its analysis ensures reliability, efficiency, and user-centric design.

Functional Requirements

Welcome Interface: The system begins with a welcoming interface that serves as the landing page. This page introduces the purpose of the platform

Basic Information Input Form: Once the user initiates the screening, the system presents a structured form to collect basic demographic and health-related information about the toddler.

Questionnaire Module: After the basic information is submitted, users are guided through the Q-Chat-10 questionnaire.

ML model-based risk prediction: The model receives the input and tells the risks based on the user input.

Functional Requirements

- Input Validation
- Risk Prediction Engine
- Result Summary Display
- Recommendation Generator
- Admin and Feedback Module

Non-Functional Requirements

- **Usability:** The application is designed to be intuitive and user-friendly. Special care is taken to make it accessible for individuals with little to no technical background.
- **Responsiveness:** All user actions, such as navigation between questions or receiving prediction results, occur within a few seconds.
- **Security:** Security is paramount, especially because the application handles sensitive information.
- **Cross-Platform Compatibility:** The application is web-based and tested to function seamlessly across common browsers like Chrome, Firefox, and Edge.

Non-Functional Requirements

- **Scalability and maintainability**
- **Reliability & Availability:**
- **Maintainability:**
- **Accessibility Compliance:**

Software Requirements

- **Operating System:** Windows 7
- **Web Browser:** Google Chrome
- **Software Development Tools:** Visual Studio Code, Jupyter Notebook
- **Programming Language:** Python 3.7
- **Web Framework:** Flask
- **Libraries/Packages:** NumPy, Pandas, Scikit-learn, XGBoost, Joblib, Jinja2
- **ML Toolkit:** scikit-learn and XGBoost
- **Development Environment:** Flask Local Server with optional virtual environments for package management

Hardware Requirements

- **Processor:** Intel Core i3
- **RAM:** 4 GB
- **Hard Disk:** 40 GB
- **Monitor:** Standard display (resolution of 1024x768)
- **Internet Connection:** Required for accessing web modules and downloading dependencies

Scope and Limitations

- Early ASD screening using web interface
- Not a diagnostic tool – for preliminary awareness only
- Requires honest and accurate user input
- Limited by diversity of training dataset
- Needs internet and device access